PRIYANKA SIWACH

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EDUCATION

Queens College, City University of New York (CUNY) *New York, NY*
Bachelor of Science in Computer Science Expected Graduation: May 2027

- GPA: 3.75/4.00, Dean's List, Honors Student
- Relevant Coursework: Data Structures & Algorithms, OOP, Cloud Computing, AI/LLM Systems, Database Systems

EXPERIENCE

AIN Ventures *New York, NY*
Software Engineer Intern June 2026 – August 2026

- Built a **Python-based sourcing agent** in Brain, AIN Ventures' internal AI platform, to re-surface promising founders below the 90-point auto-reject threshold, adding **5+ founders per day** to the outreach pipeline.
- Built a **200+ case adversarial test suite in Python** to stress-test AI guardrails, identify failure cases, and validate fixes across edge-case and malicious inputs.

Museum of Modern Art (MoMA) *New York, NY*
Cybersecurity Engineer Intern March 2024 – May 2024

- Updated and **validated identity**, photo, and badge-access records for **500+ MoMA employees,** strengthening the accuracy and integrity of the museum's physical access-control system.

PROJECTS

MiniLSM — Crash-Safe Key-Value Storage Engine *(Python, Bloom Filter, Compaction, WAL)*

- Built a LevelDB-style storage engine that lost **zero acknowledged writes** across **1,000 SIGKILL** crash trials, by designing a checksummed **write-ahead log** with torn-write recovery and atomic renames.
- Cut missing-key disk reads **99%** and raised throughput **4.5x** (5.7K → 25.4K ops/sec) with per-file **Bloom filters** and cut disk usage **54%** with k-way merge **compaction**.

ChefCoach — AI Recipe Scanner App *(React, TypeScript, Supabase, OpenAI, Capacitor, RevenueCat)*

- Launched an App Store iOS app that turns a fridge photo into diet- and allergy-filtered recipes with macros, so people cook with what they have, using GPT-4o-mini vision and RevenueCat subscriptions.
- Protected a paid AI API from abuse, blocking **30 of 40** burst requests and serving cache hits **27.8×** faster (**4.5s → 0.16s**), by moving OpenAI behind an authenticated server with **rate limits**, an **LRU cache**, and server-side quotas.

Applyfy — Job Prep & Resume Tailoring Website (applyfy.net) (*Next.js, TypeScript, Supabase, OpenAI, Stripe, Redis)*

- Built Applyfy, an AI job platform integrating OpenAI GPT-4o across **12 features**, protected by Redis-backed rate limiting, reaching **36 signups.**
- Deployed a serverless resume-parsing pipeline as code with **AWS CDK** (API Gateway, Lambda, S3, DynamoDB, WAF) that sustained **45 req/sec at 1.9s latency** under a **50-request** concurrent load test, by isolating two separate throttling layers (WAF vs. Lambda concurrency).

Collaborative Whiteboard — Real-Time Multi-User Sync *(Next.js, Node.js, WebSockets, Yjs CRDTs, Redis Pub/Sub)*

- Built a **real-time collaborative whiteboard** where multiple users edit the same board live, sustaining **100 concurrent WebSocket connections** at **100%** success in k6 load testing, with **Yjs CRDTs** and **Redis Pub/Sub**.

CERTIFICATIONS

- **AWS Certified Solutions Architect — Associate**, Amazon Web Services (AWS) — Issued June 2026
- **AWS Certified Cloud Practitioner**, Amazon Web Services (AWS) — Issued November 2025

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript

Frameworks & Libraries: React, Next.js, FastAPI, Node.js

Databases & Cloud: Redis, PostgreSQL, AWS (EC2, S3, Lambda, API Gateway, DynamoDB)

Systems Concepts: WAL, Bloom filters, compaction, LRU caching, rate limiting, CRDTs, Pub/Sub, Distributed systems

Tools: Git, GitHub Actions, pytest, Docker, CI/CD, Automated testing